# Neuromuscular Fatigue and Recovery after Heavy Resistance, Jump, and Sprint Training

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**ABSTRACT**

THOMAS, K., C. G. BROWNSTEIN, J. DENT, P. PARKER, S. GOODALL, and G. HOWATSON. Neuromuscular Fatigue and Recovery after Heavy Resistance, Jump, and Sprint Training. *Med. Sci. Sports Exerc.*, Vol. 50, No. 12, pp. 2526–2535, 2018. **Purpose**: Training methods that require maximal intensity efforts against light and heavy resistance are commonly used for athletic development. Typically, these sessions are separated by at least 48 h recovery on the assumption that such efforts elicit marked fatigue of the central nervous system (CNS), but this posit has not been well studied. The aim of the study was to assess the etiology and recovery of fatigue after heavy-resistance (strength), jump, and sprint training methods. **Methods**: Ten male athletes completed three training sessions requiring maximal efforts that varied in their loading characteristics: (i) heavy-resistance exercise (10 × 5 back squats at 80% one-repetition maximum [1RM]) (STR), (ii) jumping exercise (10 × 5 jump squats) (JUMP), and (iii) maximal sprinting (15 × 30 m) (SPR). Preparticipants, postparticipants, and 24-, 48-, and 72-h postparticipants completed a battery of tests to measure neuromuscular function using electrical stimulation of the femoral nerve, and single- and paired-pulse magnetic stimulation of the motor cortex, with evoked responses recorded from the knee extensors. Fatigue was self-reported at each time point using a visual analog scale. **Results**: Each intervention elicited fatigue that resolved by 48 (JUMP) and 72 h (STR and SPR). Decrements in muscle function (reductions in the potentiated quadriceps twitch force) persisted for 48 h after all exercise. Reductions in voluntary activation were present for 24 h after JUMP and SPRINT, and 48 h after STR. No other differences in CNS function were observed as a consequence of training. **Conclusion**: Strength, jump, and sprint training requiring repeated maximum efforts elicits fatigue that requires up to 72 h to fully resolve, but this fatigue is not primarily underpinned by decrements in CNS function. **Key Words:** NEUROPHYSIOLOGY, BRAIN, MUSCLE, TRANSCRANIAL MAGNETIC STIMULATION, CENTRAL NERVOUS SYSTEM

Athletic development in a range of sports is characterized by the application of various training means and methods to target specific adaptations. Resistance training is a key training means used by coaches and athletes to improve the strength, impulse, and speed qualities necessary for success in sports requiring movements underpinned by high force and/or velocity. The methods by which resistance training can be used in an athlete's training program can vary depending on the desired adaptive outcome. For example, to target maximum strength, coaches will typically use heavier loads (80%–95% of 1RM) with consequent slower velocities of movement (1). Conversely, to target the ability to produce high levels of force rapidly, submaximal loads are required to accrue impulse quickly (2). To train acceleration and maximum velocity running characteristics, the most effective training means is practice of sprinting itself (3). Each of these training stimuli imposes distinct demands on the athlete, but their specific consequences are not well studied or understood.

Heavy-resistance and high-velocity training methods typically require athletes to repeatedly produce maximal efforts to stimulate adaptation. An inevitable consequence of this is fatigue, a symptom or percept characterized by sensations of tiredness and weakness (4). Fatigue is a complex phenomenon and, although likely underpinned by a range of physiological and psychological mediators, an often-cited posit among athletic development professionals is that repeated maximal efforts elicit a high degree of "neuromuscular" or "central" fatigue, requiring prolonged (>48 h) recovery. Such a postulate has also recently been cited in the academic literature (5), further propagating this idea, despite a lack of peer-reviewed evidence. Neuromuscular fatigue could feasibly relate to any alteration in the physiological processes governing the central nervous system (CNS) or muscle function, but it is typically quantified by examining voluntary and artificially evoked forces during an isometric muscle action. Peripheral neuromuscular fatigue refers to impairments in

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muscle distal to the neuromuscular junction, quantified as a reduction in the resting involuntary twitch response to nervous tissue stimulation (6). Central neuromuscular fatigue is attributable to the CNS inadequately being able to activate muscle to the required level, quantified as a reduction in voluntary activation (VA) (6). Adjustments in CNS function can also be quantified via studying the evoked responses to motor cortical stimulation (7). Single- and paired-pulse magnetic stimulation of the motor cortex has been previously applied to understand acute and chronic adjustments in CNS function in response to strength training (8–12) and fatiguing single-limb (13–15) and locomotor exercise (16). In concert, the application of these techniques to study adjustments in neuromuscular function after athletic training could help explain the etiology of fatigue and aid practitioners in the appropriate scheduling of, and recovery from, different training methods.

Although decrements in neuromuscular function, particularly of the CNS, are widely considered when programming training stimuli, the evidence underpinning the idea that heavy strength and power sessions require >48 h recovery is incomplete. Previous studies recently demonstrated that heavy-resistance exercise elicited greater acute reductions in voluntary force than a similar low-resistance, high-velocity "power" session (17), and that these heavy-resistance exercise induced decrements persisted at 24 h postexercise in elite athletes (18). Bartolomei et al. (19) recently demonstrated greater and more prolonged strength and jump performance impairments after "hypertrophy" style training (higher volume, lower load, and shorter rest periods) compared with a training stimulus targeting strength development (lower volume, higher intensity, and longer rest periods). Collectively, these findings suggest that the acute and prolonged adjustments underpinning the fatigue experienced after resistance exercise vary between training methods, but these studies were limited by both the range of outcome measures studied, and/or a limited profile of the time-course recovery of neuromuscular function. Further study is warranted to comprehensively assess the acute and prolonged neuromuscular adjustments induced by the typical training means and methods commonly used in the physical preparation of athletes. Such information will be of high value to practitioners when prescribing training stimuli.

The aim of the study was to assess the etiology and recovery of neuromuscular fatigue in response to heavy-resistance, jumping, and sprinting exercise. It was hypothesized that the maximal nature of all exercise interventions would induce marked neuromuscular fatigue that would require >48 h to resolve and that the time course of recovery would be similar between interventions.

## METHODS

### Participants

Ten male participants (age, 21 ± 2 yr; stature, 1.82 ± 0.05 m; mass, 85 ± 12 kg) gave their written, informed consent to participate in the study, which was approved by the Northumbria University Faculty of Health and Life Sciences Ethics Committee. All participants had >3 yr history of training experience using resistance and maximal speed methods and were currently competing in intermittent ($n$ = 6) or track and field ($n$ = 4) sports at university or national standard.

### Design

Participants initially visited the laboratory on two separate occasions for preliminary assessments and to habituate to the measurement tools of the study. Subsequent to this participants completed three experimental trials, each spanning four consecutive days and separated by 1 wk, in a randomized, counterbalanced order. On the first day of each experimental trial, participants completed one of three interventions as follows: (i) a heavy-resistance exercise session consisting of repeated sets of back squats (STR); (ii) a low-load, high-velocity exercise session consisting of repeated sets of jump squats (JUMP); and (iii) a maximal speed training session consisting of repeated 30 m sprints (SPR). Preexercise, immediately postexercise, and at 24, 48, and 72 h postexercise, a battery of assessments to measure fatigue and neuromuscular function were administered. Before all visits, participants were instructed to refrain from caffeine (24 h) and alcohol (48 h) and to arrive 2 h postprandial in a fully rested, hydrated state. Participants were also instructed not to perform any exercise other than that required by the study for the duration of their participation. To account for any potential detraining-induced changes in physical fitness, a "refresh" session consisting of maintenance loads for the physical qualities under study was used between experimental trials. An overview of the experimental trials can be viewed in Supplemental Digital Content 1 (see Figure, Supplemental Digital Content 1, Schematic of experimental protocol, http://links.lww.com/MSS/B357).

### Procedures

**Practice trial.** Before the experimental trials, participants visited the laboratory on two occasions for habituation to the measurement tools of the study (on both visits) and an assessment of 1RM back squat strength or jump squat performance (on separate visits). Before all exercise (practice and experimental trials), participants completed a structured 10-min warm-up, which incorporated jogging, dynamic flexibility movements, mobility exercises specific to squatting, jumping, and sprinting, and 3 × 30 m progressive strides at 70%, 80%, and 90% of perceived maximum sprint speed. For the assessment of maximum isoinertial strength, participants first completed warm-up sets of three to five repetitions of back squats (high bar position), beginning with an unloaded barbell and progressing to 50%, 70%, 80%, and 90% of their estimated 1RM. The load on the bar was then incremented by 2%–5% until participants could not complete one repetition. The technical execution of each lift required participants to descend under control (2-s tempo) to a depth where the femur was parallel to the floor. Participants





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then immediately reversed the movement and were instructed to maximally accelerate the bar during the concentric phase. A repetition was deemed unsuccessful if participants could not complete the concentric phase in ≤2 s. Maximum isoinertial strength was 126 ± 14 kg, or 150% ± 15% body mass. For jump squats, participants completed vertical jumps for maximum height, beginning with body mass (plus a wooden dowel) and incrementing by 5 kg; the first increment was achieved by replacing the dowel with a lightweight training barbell with a mass of 5 kg. Each repetition required participants to squat to a self-selected depth (approximating a half squat) and jump for maximum height. Jump height was recorded using photoelectronic timing gates (Optojump Next, Microgate, Milan, Italy) for two to three efforts at each load. When participants were unable to maintain performance within 5% of their unloaded jump height because of added resistance, the test was terminated, and the highest applied load where squat jump height was maintained was used for experimental trials (mean, SD 10 ± 5 kg, with a range of 0 to 20 kg, additional load).

**Experimental trials: exercise intervention.** On the first day of each experimental trial, subsequent to pretest assessment, participants completed one of three exercise prescriptions: (i) heavy-resistance training consisting of 10 × 5 repetitions of the high bar back squat at 80% 1RM, with 3 min recovery (STR); (ii) 10 × 5 repetitions of jump squats, with 3 min recovery (JUMP); and (iii) 15 × 30 m maximum sprints, with 2 min recovery (SPR). For STR and JUMP, participants were encouraged to maximally accelerate the load, and the velocity of each repetition was monitored using a wearable linear position transducer (PushBand, Heap Analytics, Toronto, Canada). For SPR, participants began each sprint 0.5 m behind the first timing gate and were encouraged to sprint maximally through the timing gate at 30 m. Each sprint was measured using photocell technology (TC Timing system; Brower Timing Systems, Draper, UT). For all trials, participants were provided feedback on the execution of each repetition to promote a maximum effort. Posttraining, participants were asked for a whole trial session RPE using the 0–10 category ratio scale. Although it was impossible to equate training load between the experimental trials, the configurations for STR, JUMP, and SPR were designed in consultation with experienced strength and conditioning coaches to represent a "heavy" stimulus for the physical quality under stress and were similar in duration (approximately 45 min, including the standardized warm-up).

**Experimental trials: outcome measures.** On each occasion, participants completed a battery of assessments to measure fatigue and neuromuscular function. All outcome measures were assessed preexercise, postexercise, and at 24, 48, and 72 h postexercise, unless otherwise stated.

**Visual analog scales and creatine kinase.** Upon arrival, and postexercise after assessment of neuromuscular function, participants completed visual analog scales (VAS, 100 mm scale) to record fatigue and perceptions of muscle soreness. For fatigue, the VAS was anchored with the verbal descriptors "not fatigued at all" to "extremely fatigued"; participants were asked to rate their general feeling of "fatigue, tiredness, weakness, and lethargy." For muscle soreness, the VAS was anchored with "no soreness" to "extremely sore"; participants preceded their rating with three repetitions of a body weight squat and were asked to rate their "muscle soreness and pain." Subsequent to this, fingertip samples of capillary blood were obtained and immediately assayed for creatine kinase (CK) concentration (Reflotron, Roche Diagnostics, Germany).

**Assessment of neuromuscular function.** The evoked force and EMG responses of the rectus femoris (RF) to transcranial magnetic stimulation (TMS) of the primary motor cortex, and electrical stimulation of the femoral nerve, were used to assess neuromuscular fatigue, corticospinal excitability, and the status of inhibitory intracortical networks. The assessment of neuromuscular function took place subsequent to perceptual assessments and capillary blood sampling at all time points except for postexercise, where it was conducted first to capture the extent of neuromuscular fatigue elicited by the exercise intervention.

A calibrated load cell (MuscleLab force sensor 300; Ergotest Technology, Langesund, Norway) recorded muscle force (N) during an isometric maximal voluntary contraction (iMVC) of the knee extensors. During contractions, participants sat with hips and knees at 90° flexion, with a load cell fixed to a custom-built chair and attached to the participants right leg, superior to the ankle malleoli, with a noncompliant cuff. Electrical activity from the RF and bicep femoris were recorded from surface electrodes (Ag/AgCl; Kendall H87PG/F, Covidien, Mansfield, MA) placed 2 cm apart over the belly of each muscle, with a reference electrode placed on the patella. Electrode placement was marked with indelible ink to ensure consistent placement throughout the study, with the areas cleaned and shaved before electrode placement. The electrodes recorded the root mean square (RMS) amplitude for submaximal and maximal voluntary contractions, the compound muscle action potential (M-wave) from the electrical stimulation of the femoral nerve, and the motor evoked potential (MEP) elicited by TMS. Signals were amplified: gain ×1000 for EMG and ×300 for force (CED 1902; Cambridge Electronic Design, Cambridge, UK), band-pass filtered (EMG only: 20–2000 Hz), digitized (4 kHz; CED 1401, Cambridge Electronic Design), and analyzed offline. Further details on these methods are provided below.

**Motor nerve stimulation.** Motor nerve stimulation was used for the measurement of contractile function, muscle membrane excitability, and VA. Single electrical stimuli were administered using square wave pulses (200 $\mu$s) via a constant-current stimulator (DS7AH; Digitimer Ltd., Hertfordshire, UK) using self-adhesive surface electrodes (Nidd Valley Medical Ltd., North Yorkshire, UK). Electrical stimuli were first administered to the motor nerve at rest in 20-mA stepwise increments from 20 mA until the maximum quadriceps twitch amplitude ($Q_{tw}$, N) and muscle compound action potential ($M_{max}$, mV) were elicited. To ensure a

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consistent, supramaximal stimulus and account for any activity-induced changes in axonal excitability, the resulting stimulation intensity was increased by 30% for all subsequent stimulus. The peak-to-peak amplitude and area of the electrically evoked maximal compound action potential ($M_{max}$) was used as a measure of membrane excitability. Participants subsequently completed six iMVCs (3- to 5-s duration) of the knee extensors, separated by 60-s rest. For the final three iMVCs, electrical stimuli were delivered during and 2 s postcontraction to assess VA and potentiated quadriceps twitch force ($Q_{tw,pot}$), respectively.

**Motor cortical stimulation.** Single- and paired-pulse TMS of 1-ms duration were delivered using a concave double cone coil using two linked monopulse magnetic stimulators (Magstim 200; The Magstim Company Ltd., Whitland, UK). The junction of the double cone coil was aligned tangentially to the sagittal plane, with its center 1–2 cm to the left of the vertex. The optimal coil placement was determined at the start of each trial as the position that elicited the largest MEP in the RF, with a concomitant small MEP in the bicep femoris. The position was marked with indelible ink for consistent placement during subsequent trials. The stimulator intensity was based on active motor threshold (AMT) measured during a 10% iMVC. To determine AMT, the stimulator intensity was increased in 5% steps beginning at 35% of stimulator output until a consistent MEP with peak-to-peak amplitudes of >200 $\mu$V was found. Thereafter, stimulus intensity was reduced in 1% step until an MEP of >200 $\mu$V was found in 50% of stimulations.

**Corticospinal excitability and short-interval intracortical inhibition.** Once AMT was established, the stimulator intensities required to assess the MEP response to varying TMS intensities (stimulus–response curve) were determined to assess corticospinal excitability. Participants held a submaximal voluntary contraction (10% iMVC) with one set of five stimuli delivered at each of 90%, 100%, 110%, 120%, 130%, 140%, 150%, and 160% of AMT in a randomized and counterbalanced order, with 4–6 s between each stimuli and 15 s between each set. For short-interval intracortical inhibition (SICI), 10 single and 10 paired-pulse TMS stimuli were administered in two sets of 10 stimuli during a 10% iMVC, for measurement of unconditioned and conditioned MEP amplitude, respectively. Paired-pulse TMS consisted of a subthreshold conditioning pulse at 70% of AMT, and a suprathreshold test pulse at 120% AMT, with an interstimulus interval of 2 ms. Single and paired pulses ($\times$10 each) were delivered in a predetermined randomized order, with 4–6 s between each stimulation and a short rest between each set. This assessment was conducted preexercise, and at 24-h intervals thereafter until 72-h post.

**Voluntary activation with TMS.** Single pulse TMS was delivered during brief (3–5 s) contractions at 100%, 75%, and 50% iMVC, separated by 5 s of rest, for determination of VA with TMS ($VA_{TMS}$). This procedure was repeated three times with 15-s rest between each set. The stimulation intensity was set at the stimulator output that elicited the maximum superimposed twitch force (SIT) during a 50% iMVC. The SIT force elicited from contractions at 100%, 75%, and 50% were used to determine $VA_{TMS}$ (see data analysis section for details).

**Experimental trials: "refresh session."** On the final day of each experimental trial, after all outcome measures had been completed, a "refresh" session designed to maintain the physical qualities under study over the course of the experimental period was used. This consisted of a low-volume, high-intensity stimulus for each physical quality in a single session (3 $\times$ 5 sets of back squats at 80% 1RM, 3 $\times$ 5 maximal effort jump squats, and 3 $\times$ 30 m maximal effort sprints). Previous research has demonstrated that strength qualities can be adequately maintained for prolonged periods using low doses provided the intensity of exercise remains close to maximal (20,21).

## Data Analysis

Voluntary activation assessed through the interpolated twitch technique (22) was quantified by comparing the amplitude of the superimposed twitch force to the potentiated twitch (100 Hz) delivered 2 s after the iMVC at rest using the following equation: motor point VA (%) = $[1 - (SIT/Q_{tw,\ pot}) \times 100]$. Voluntary activation using TMS ($VA_{TMS}$) was assessed during contractions at 50%, 75%, and 100% iMVC using linear regression of the superimposed twitch force evoked by TMS (23), with the regression analysis confirming a linear relationship at each time-point ($r^2$ range = 0.89 ± 0.03 to 0.95 ± 0.04). The estimated resting twitch (ERT) was calculated as the $y$-intercept of the linear regression between the mean amplitude of the SIT force evoked by TMS at each contraction intensity. Subsequently, $VA_{TMS}$ was quantified using the equation $[1 - (SIT/ERT) \times 100]$. To quantify SICI, the ratio of the average conditioned paired-pulse MEP was expressed relative to the average unconditioned MEP at 120% AMT. Recruitment curves were constructed by plotting the TMS stimulation intensity relative to AMT against the MEP amplitude averaged from the five stimulations at each intensity, expressed relative to $M_{max}$. The ratio of the MEP amplitude to the maximum M-wave was used as an index of corticospinal excitability. To provide a summary measure of corticospinal excitability, the summated area under the stimulus–response curve was calculated for each participant at each time point using the trapezoid integration method (24). The $EMG_{RMS}$ amplitude and average force was calculated in the 80 ms before each TMS to ensure a similar level of background muscle activity was present during the stimulus–response curve and SICI measurements. The peak-to-peak amplitude of evoked MEP and $M_{max}$ were measured offline.

## Statistical Analysis

Data are presented as mean ± SD. To ascertain the time-course recovery of neuromuscular fatigue, within-trial, one-way repeated-measures ANOVA across time was used for STR, JUMP, and SPR data. Significant main effects were followed up with Dunnett's multiple comparison procedure,





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TABLE 1. Within-trial differences in fatigue and perceptions of muscle soreness measured using visual analog scales (100-mm scale), and CK, measured preexercise and in the 72 h postexercise strength, jump, and sprint training.

| | Strength | Jump | Sprint |
|---|---|---|---|
| Fatigue (mm) | | | |
| Preexercise | 16 ± 13 | 14 ± 11 | 16 ± 6 |
| Postexercise | 63 ± 16* | 44 ± 15* | 56 ± 11* |
| 24 h | 52 ± 19* | 28 ± 15* | 51 ± 21* |
| 48 h | 56 ± 19* | 30 ± 16 | 40 ± 16* |
| 72 h | 26 ± 16 | 22 ± 17 | 27 ± 13 |
| Muscle soreness (mm) | | | |
| Preexercise | 16 ± 13 | 15 ± 13 | 18 ± 9 |
| Postexercise | 47 ± 22* | 34 ± 10* | 39 ± 17* |
| 24 h | 61 ± 22* | 25 ± 11* | 68 ± 17* |
| 48 h | 63 ± 23* | 33 ± 21* | 52 ± 21* |
| 72 h | 40 ± 29 | 20 ± 21 | 31 ± 18 |
| CK ($IU \cdot L^{-1}$) | | | |
| Preexercise | 185 ± 98 | 253 ± 114 | 265 ± 142 |
| 24 h | 863 ± 659* | 569 ± 340* | 946 ± 531* |
| 48 h | 733 ± 673 | 547 ± 328* | 622 ± 357* |
| 72 h | 440 ± 333 | 356 ± 205 | 484 ± 270* |

Values are presented as mean ± SD.
*Significant within-trial difference from pretest score.

with the preexercise score used as the control category. To assess between-trial differences in the magnitude of neuromuscular fatigue induced by STR, JUMP, and SPR, two-way (trial–time) factorial repeated-measures ANOVA analysis was used. As baseline scores did not differ between trials for any outcome measure, significant trial–time interaction effects were followed up with one-way repeated-measures ANOVA and *post hoc* Tukey-adjusted pairwise comparisons at each time point to locate statistically significant between-trial differences. The assumptions underpinning these statistical procedures were verified as per the guidelines outlined by Newell et al. (25). Data were analyzed using GraphPad Prism (version 7; GraphPad Software Inc., La Jolla, CA). Statistical significance was accepted at $P < 0.05$.

## RESULTS

**Exercise responses.** All participants successfully completed the prescribed training interventions. For STR, the load lifted was 101 ± 11 kg. Repetition velocity decreased from 0.53 $m \cdot s^{-1}$ in set 1 to 0.44 $m \cdot s^{-1}$ in set 10 ($P < 0.05$), with a best of 0.54 ± 0.07 $m \cdot s^{-1}$ and worst of 0.41 ± 0.07 $m \cdot s^{-1}$ independent of set. Session RPE averaged 8 ± 2 for STR. For JUMP, mean repetition velocity was successfully maintained throughout the exercise (1.61 ± 0.17 $m \cdot s^{-1}$ in set 1 vs 1.56 ± 0.14 $m \cdot s^{-1}$ in set 10, $P = 0.31$, best score of 1.69 ± 0.11 $m \cdot s^{-1}$, worst of 1.48 ± 0.10 $m \cdot s^{-1}$), and session RPE was lower (5 ± 1) than STR ($P = 0.001$). For SPR, 40-m sprint time declined from 4.40 ± 0.14 s in set 1 to 4.55 ± 0.22 s in set 15 ($P = 0.04$), with a fastest sprint of 4.36 ± 0.16 s and a slowest of 4.61 ± 0.24 s. Session RPE after SPR (6 ± 2) was not different to STR ($P = 0.18$) or JUMP ($P = 0.33$).

**Perceived fatigue and muscle damage responses.** All exercise interventions elicited significant perceived fatigue (Table 1) that persisted for 48 h after STR (48 h, $P = 0.002$) and SPR training (48 h, $P = 0.008$) and 24 h after JUMP training (24 h, $P = 0.02$). Between trials, both STR and SPR training resulted in greater perceived fatigue than JUMP training for up to 48 h (Fig. 1A). Similar patterns were also evident for perceptions of muscle soreness; all training resulted in increases in muscle soreness that were different to baseline for 48 h, and between trials, both STR (for up to 72 h, $P = 0.0006$) and SPR (for up to 48 h, $P = 0.0008$) elicited a greater magnitude of soreness in comparison with JUMP (Fig. 1B). CK peaked at 24 h in all trials and was different to baseline for 24, 48, and 72 h for STR, JUMP, and SPR,

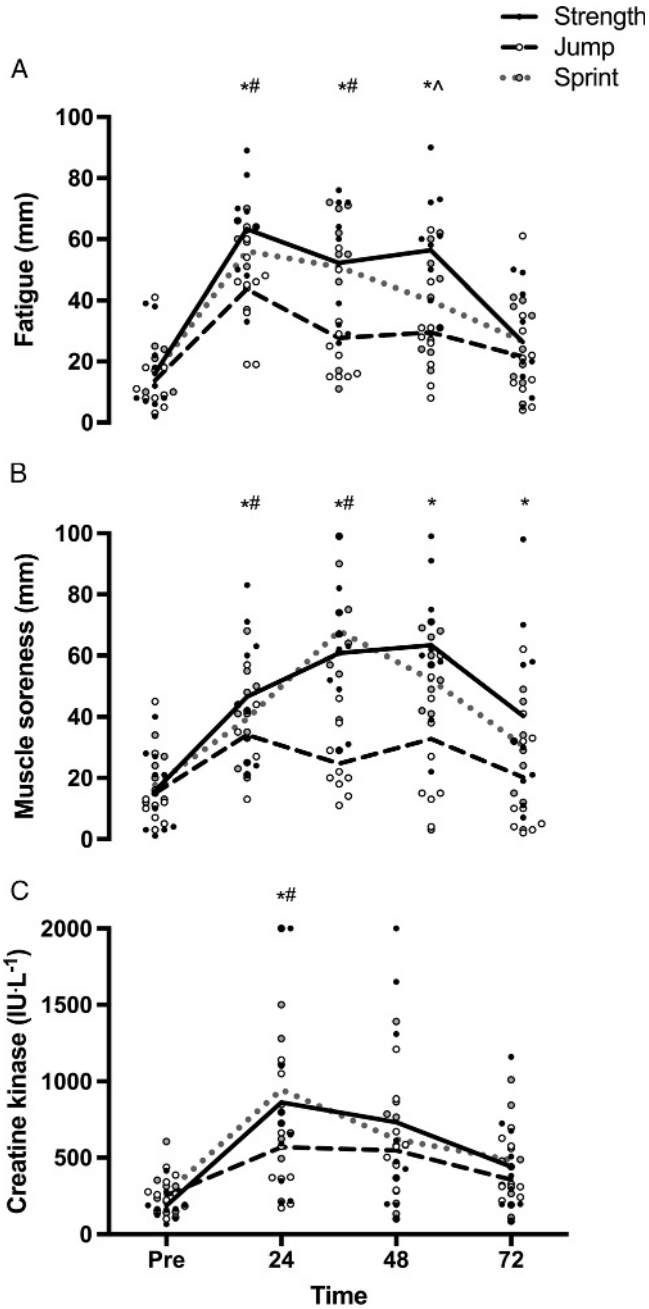


**FIGURE 1—Between-trial differences in fatigue (A), muscle soreness (B), and CK (C) measured preexercise, postexercise, and 24, 48, and 72 h postexercise strength, jump, and sprint training. Between-trial differences: * difference between strength and jump; # difference between jump and sprint; ^ difference between strength and sprint (all $P > 0.05$). Individual responses are plotted, with lines representing the mean score.**

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TABLE 2. Within-trial differences in isometric maximum voluntary contraction strength and measures of neuromuscular fatigue preexercise, postexercise, and 24, 48, and 72 h postexercise strength, jump, and sprint training.

| | Strength | Jump | Sprint |
|---|---|---|---|
| iMVC (N) | | | |
| Preexercise | 691 ± 78 | 693 ± 78 | 693 ± 74 |
| Postexercise | 548 ± 61* | 611 ± 52* | 614 ± 66* |
| 24 h | 600 ± 78* | 630 ± 63* | 627 ± 72* |
| 48 h | 637 ± 90* | 644 ± 77* | 650 ± 83* |
| 72 h | 678 ± 102 | 686 ± 77 | 682 ± 78 |
| VA (%) | | | |
| Preexercise | 92.4 ± 2.9 | 92.2 ± 2.7 | 92.3 ± 2.6 |
| Postexercise | 84.5 ± 5.8* | 84.8 ± 6.1* | 86.1 ± 4.7* |
| 24 h | 87.6 ± 3.3* | 89.4 ± 3.8* | 88.1 ± 3.5* |
| 48 h | 88.2 ± 4.4* | 89.9 ± 3.8 | 89.5 ± 3.3 |
| 72 h | 91.4 ± 3.2 | 92.0 ± 3.2 | 91.1 ± 2.9 |
| $VA_{TMS}$ (%) | | | |
| Preexercise | 94.7 ± 2.5 | 94.0 ± 2.4 | 94.2 ± 2.0 |
| Postexercise | 86.9 ± 5.7* | 89.1 ± 4.7* | 87.7 ± 5.3* |
| 24 h | 90.7 ± 5.7* | 91.5 ± 5.1 | 91.2 ± 4.0* |
| 48 h | 92.8 ± 4.1 | 93.3 ± 4.4 | 92.5 ± 3.4 |
| 72 h | 93.2 ± 3.5 | 94.5 ± 3.3 | 94.2 ± 2.0 |
| $Q_{tw,pot}$ (N) | | | |
| Preexercise | 209 ± 33 | 203 ± 31 | 207 ± 36 |
| Postexercise | 129 ± 14* | 151 ± 19* | 156 ± 27* |
| 24 h | 186 ± 30* | 187 ± 29* | 184 ± 30* |
| 48 h | 196 ± 34* | 192 ± 31* | 198 ± 36* |
| 72 h | 204 ± 31 | 204 ± 33 | 203 ± 38 |

Values are presented as mean ± SD.
*Significant within-trial difference from pretest score.
iMVC, isometric maximum voluntary contraction; $VA_{TMS}$, VA measured with transcranial magnetic stimulation; $Q_{tw,pot}$, quadriceps potentiated twitch force.

respectively (Table 1). Between trials, CK was lower at 24 h in JUMP compared with both STR ($P = 0.001$) and SPR ($P = 0.002$) (Fig. 1C).

**Neuromuscular fatigue.** All exercise interventions resulted in declines in iMVC force that took until 72 h to fully resolve in all trials (Table 2). The magnitude of the reduction in iMVC force immediately postexercise was higher after STR compared with JUMP ($P < 0.001$) and SPR ($P < 0.001$), a difference that persisted at 24 h ($P = 0.02$ and 0.05, respectively; Fig. 2A). Reductions in VA were also evident immediately postexercise for all trials and persisted for 48 h after STR ($P = 0.004$) and 24 h after JUMP ($P = 0.015$) and SPR ($P = 0.023$, Table 2). Significant reductions in $VA_{TMS}$ were also evident postexercise in all trials (all $P < 0.05$) but returned to baseline quicker than VA, by 48 h in STR and 24 h in JUMP and SPR (Table 2). The magnitude of reductions in VA, measured with both motor nerve and motor cortical stimulation, was not different between exercise interventions (Fig. 2B and C). All trials resulted in reductions in $Q_{tw,pot}$, that took 72 h to fully resolve (Table 2). Between trials, there were larger reductions in $Q_{tw,pot}$ immediately–post-STR compared with both JUMP and SPR (both $P < 0.001$), with no differences between trials thereafter (Fig. 2D).

**Corticospinal excitability and SICI.** Exercise resulted in no modulation of corticospinal excitability (Fig. 3, stimulus–response curves) or SICI (Fig. 4), both within and between trials (all $P > 0.05$). The $EMG_{RMS}$ was also not different within and between trials [see Table, Supplemental Digital Content 2, Surface electromyographic responses to transcranial magnetic stimulation and electrical

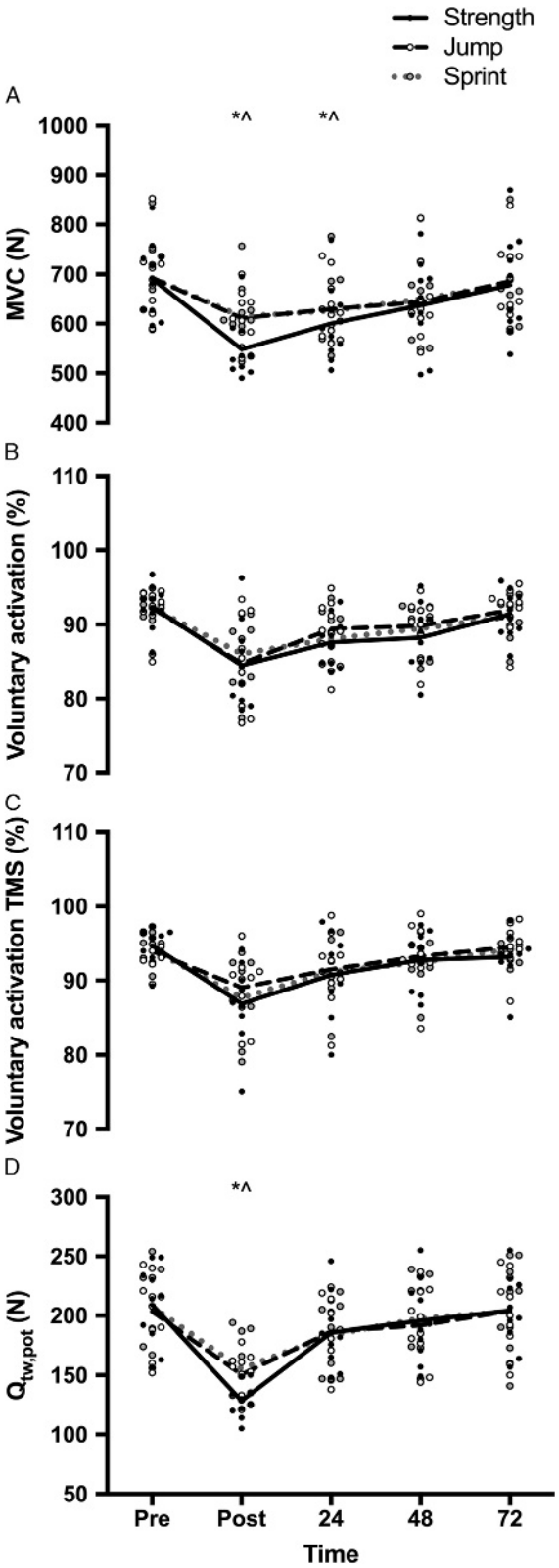


**FIGURE 2—Between-trial differences in isometric maximum voluntary contraction force (A), VA measured with motor nerve (B) and motor cortical (C) stimulation, and quadriceps potentiated twitch force (D). Between-trial differences: * difference between strength and jump; # difference between jump and sprint; ^ difference between strength and sprint (all $P > 0.05$). Individual responses are plotted, with lines representing the mean score.**

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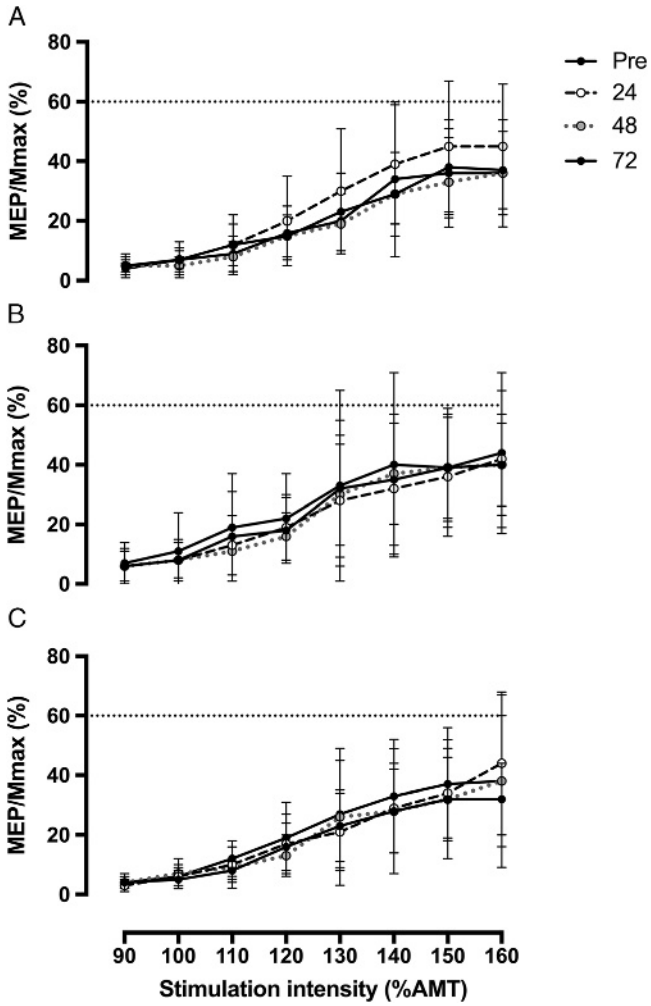


**FIGURE 3—MEP (expressed relative to maximum M-wave) stimulus–response curves measured above and below AMT (100%) preexercise, and 24, 48, and 72 h postexercise strength (A), jump (B), and sprint (C) training. Values are presented as mean ± SD. A reference line is included at 60% to assist comparison between trials.**

stimulation of the femoral nerve at rest and during contraction ($n$ = 10) pre- and at 24, 48 and 72 h post strength, jump and sprint training, http://links.lww.com/MSS/B358]. For a full list of surface EMG responses to TMS and electrical stimulation, please see Supplemental Digital Content 2 [see Table, Supplemental Digital Content 2, Surface electromyographic responses to transcranial magnetic stimulation and electrical stimulation of the femoral nerve at rest and during contraction ($n$ = 10) pre- and at 24, 48 and 72 h post strength, jump and sprint training, http://links.lww.com/MSS/B358].

## DISCUSSION

The aim of the study was to assess the effect of strength, jump, and sprint training, performed with maximal intent, on the etiology and time course of neuromuscular fatigue and recovery. In accordance with our hypothesis, all training stimuli resulted in neuromuscular adjustments that took up to 72 h to fully resolve. For twitch force, indicative of peripheral fatigue, strength training resulted in larger postexercise reductions compared with jump and sprint training, but the time-course recovery was similar thereafter, with marked decrements still evident at 48 h postexercise in all trials. Reductions in VA, an indicator of central fatigue, persisted for 24 h after jump and sprint training and 48 h after strength training, with no difference between trials in the magnitude of these reductions. Measures of CNS responsiveness and inhibition were not modulated in response to the training stimuli at any time point. Perceptual indicators of fatigue and soreness followed a similar time course of recovery to measures of neuromuscular function, requiring up to 72 h to return to baseline, with a tendency for jump training to be less fatiguing compared with strength and sprint training. Collectively, these data indicate that maximal intent, relatively high volume, strength, jump, and sprint training methods elicit neuromuscular fatigue, mediated by both central and peripheral mechanisms, that requires up to 72 h to fully resolve.

**Time course of recovery of neuromuscular fatigue after training.** An often-cited posit in strength and conditioning is the idea that training methods performed with maximal intent, such as those studied here, result in central fatigue, or are CNS intensive, and require 48–72 h recovery before similarly intense stimuli are imposed (26,5,27). To date, however, the formal study of neuromuscular fatigue in the days posttraining has been limited (19,17,18,28,29). Here we show that strength, jump, and sprint training elicits marked neuromuscular central and peripheral fatigue that can require up to 72 h to fully resolve, which provides some support to these previous assertions. The capacity to produce voluntary force was impaired for 48 h after all training, with decrements in MVC force of 8%, 7%, and 6% on average for strength, jump, and sprint training.

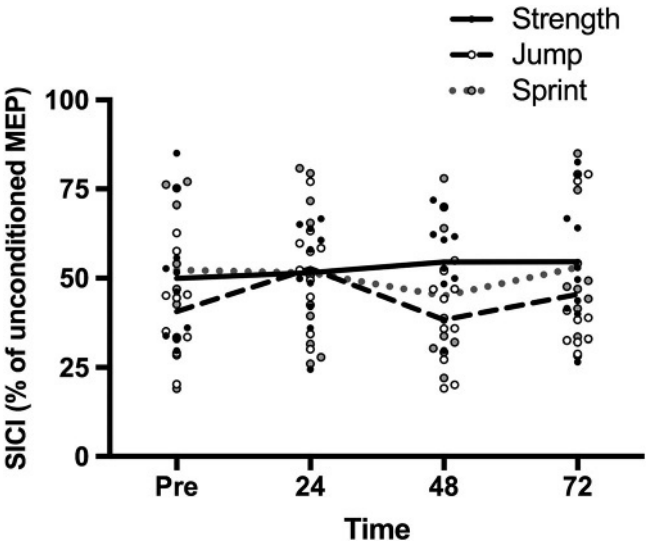


**FIGURE 4—SICI expressed as the ratio between conditioned and unconditioned MEP preexercise, and 24, 48, and 72 h postexercise strength, jump, and sprint training. Individual responses are plotted, with lines representing the mean score.**





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Similarly, twitch force was reduced compared with baseline for 48 h in all trials, indicating a prolonged decrement in muscle function, with values remaining depressed by 5%–6% on average at 48 h. Reductions in VA persisted for 48 h after strength training and 24 h after jump and sprint training, suggesting heavy-resistance training elicited more prolonged central fatigue than the other methods studied. At the 48-h time point, the decrement in VA averaged 5%, 2%, and 3% for strength, jump, and sprint training, respectively. Collectively, these data suggest that neuromuscular fatigue after training methods that emphasize maximal intent is persistent and multifactorial. This underscores the need for appropriate recovery between such sessions, alongside interventions that address the multifactorial nature of fatigue. The data also provide some support to the assertion that training sessions that emphasize maximal intent should be separated by at least 48 h if peak performance is a priority, as the majority of variables under study took 72 h to fully resolve.

**"Central" fatigue after training.** Fatigue of the CNS is often implicated as a primary consideration after training modes that emphasize maximal intent and recent reviews have called for an increased emphasis on the recovery of central and "brain" fatigue after exercise (30,31). However, the formal study, and precise definition, of what constitutes central fatigue is limited. Here we specifically measured central fatigue as a reduction in the ability of the CNS to activate skeletal muscle. This activation deficit was evident posttraining for up to 24 h after jump and sprint training and up to 48 h after heavy-resistance training. We also measured variables purported to reflect CNS excitability and inhibition, but these did not modulate with training. By contrast, the capacity to produce voluntary force was impaired for 48 h in all trials, decrements in muscle function (indicative of peripheral fatigue) persisted for 48 h in all trials, and sensory perceptions of fatigue and soreness persisted for 48–72 h post. The magnitude of central fatigue was also modest, with VA returning to within 5% of baseline in the majority of cases ($n$ = 6, 8, and 6, respectively, for strength, jump, and sprint training) by 24 h post. In addition, the magnitude of the decrement posttrial was similar to that previously observed in our laboratoryfor prolonged cycling exercise (32,33), repeated-sprint exercise (34), and simulated intermittent-sprint exercise (35). The recovery of central neuromuscular fatigue in the days postexercise was also similar to that observed after simulated intermittent-sprint exercise (35). Therefore, the idea that recovery of the CNS should be prioritized after methods of training that emphasize maximal intent is debatable, but perhaps it simply reflects an imprecise definition of terms. Fatigue is a symptom, or percept, characterized by sensations of tiredness and weakness (4), underpinned by a myriad of physiological and psychological mechanisms; what is commonly perceived as central fatigue by athletes and coaches is likely more accurately interpreted as fatigue *per se*. That is, the feelings of tiredness and weakness that athletes experience in the days postexercise are likely underpinned by a range of mechanisms relating to both central and peripheral function, and not primarily attributable to "CNS" fatigue. A caveat to this conclusion is the acknowledgment that our ability to measure aspects of CNS function, and thus infer the impact of exercise, is limited by the available measurement tools. For example, even the most widely acknowledged measure of central fatigue – a reduction in VA of skeletal muscle – has questionable validity (36). This notwithstanding, our data suggest that the fatigue experienced after the training methods under study is multifactorial and not primarily underpinned by central mechanisms.

**Differential effect of strength, jump, and sprint training.** Several differences were observed between trials that indicated the jumping training stimulus elicited less fatigue and took less time to recover from. These included differential effects on iMVC and twitch force, the CK response, and perceptions of fatigue and muscle soreness, in comparison with heavy-resistance exercise and sprinting. However, whether these differences could be primarily attributed to differences in the force–velocity requirements of the differing sessions is debatable. Both the heavy-resistance (back squat to parallel depth) and the sprinting stimuli required greater displacement of load (external or body mass) in comparison with power training (jumping from a half squat). The ostensibly increased work required during STR and SPR (and associated metabolic demand) and the increased potential for muscle damage at longer muscle lengths could explain the differences observed between trials independent of differences in the force–velocity demands of the exercise. Equating the training stimulus between trials is an impossible endeavor, and therefore any between-trial comparisons should be interpreted with caution. However, the relatively lower stress and quicker recovery observed after jumping compared with heavy-resistance training is not without precedent. Howatson et al. (18) previously observed strength training (consisting of 4 × 5 heavy back squat, split squat, and push press) elicited reductions in iMVC for up to 24 h, whereas the same session conducted with lower loads and higher repetition velocities elicited no reduction in iMVC. In addition, Linnamo et al. (29) previously demonstrated a higher degree of acute neuromuscular fatigue following heavy load versus light load "explosive" bilateral leg extension resistance training. These previous data, and the current study, indicate that training methods that emphasize the ability to generate impulse to accelerate relatively light loads might require less recovery time than heavy-resistance or maximal sprint training, a finding that has implications for the scheduling of such activities.

**Corticospinal excitability and short intracortical inhibition.** There were no discernible adjustments in corticospinal excitability nor short intracortical inhibition at any time point in response to all exercise interventions. Corticospinal excitability has been shown to modulate acutely with single limb fatiguing exercise (13–15) and ballistic isometric exercise (9), and chronically after single limb (8,12) and whole body (10) resistance training programs. Short intracortical inhibition has similarly been demonstrated





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to be modulated after a period of resistance training (10), and acutely during locomotor exercise (16) and after fatiguing isometric knee extensor exercise (37). Of importance, these acute adjustments seem to quickly resolve upon exercise cessation (37,16); this could explain why, in the present study, we did not observe any differences postexercise as the measurement of these variables was delayed in comparison with previous work. The finding that neither corticospinal excitability nor short intracortical inhibition were modulated with recovery in the days postexercise concurs with previous studies from our laboratory studying the etiology and recovery of neuromuscular fatigue after simulated and competitive intermittent-sprint exercise (38,35). Thus, although measures of CNS excitability and inhibition might be modulated during and immediately postexercise, or chronically in response to longer-term training, they do not systematically differ from baseline in the days after fatiguing exercise.

In addition to an inability to match training stimuli between trials, the ecological validity of both the imposed sessions and the measurement protocols could also be questioned. Considering the primary variables under study (i.e., indicators of neuromuscular fatigue), we deliberately chose to study a high volume of exercise for each training stimulus and limited each to a single exercise that required a significant contribution from the quadriceps muscle group, and where possible were biomechanically similar (e.g., back squats vs jump squats). For these reasons, the applicability of the results to regular athletic development training, which typically involves lower volumes and higher variation of exercises within sessions, is questionable. There are of course unlimited configurations of exercise selection, sets, repetitions, and recovery durations that could be manipulated, and consequently any decisions on the exercise intervention used in a study of this nature could be questioned. In addition, adjustments in neuromuscular function as a consequence of exercise were studied during single-limb, isometric knee extensor muscle actions. This assessment setup is required to measure neuromuscular fatigue; however, these adjustments might not fully reflect decrements in the type of dynamic knee extensor function required of the exercise modes under study and athletic performance more generally. These limitations notwithstanding, the data do provide new information on the nature of fatigue and recovery after resistance and speed training, an area of research that is understudied and in need of further investigation.

In conclusion, this study has demonstrated that training methods requiring repeated maximal intensity efforts elicit marked neuromuscular fatigue that requires up to 72 h to fully resolve. The observed neuromuscular fatigue was of both a central and peripheral origin, with a faster recovery of central, compared with peripheral, neuromuscular fatigue. The data provide partial support for the idea that training methods that emphasize maximal intent to express force or velocity should be separated by at least 48 h, but the recovery of CNS function is not necessarily the primary aim of this period. Rather, the residual fatigue experienced by athletes after such training is multifactorial, and thus development of appropriate monitoring and rest/recovery strategies that reflect this is warranted. Further research is required to further probe the consequences of maximal intensity training using novel measurement tools and stimuli that more accurately reflect the day-to-day practice of different athletic groups.

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The authors have no conflict of interest to declare. The results of the study do not constitute endorsement by the American College of Sports Medicine. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation.

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